

3M Window Film

Bangchak project

April 23, 2025

Benefit of Applying Window Film



3เอ็ม ฟิล์มอาคาร
ผู้ให้กำเนิดฟิล์มกรองแสงรายแรกของโลก

Solution for Energy Efficiency

Solar Control Films

Window films are a type of window treatment applied to windows and glass surfaces, designed to reduce the amount of solar heat transmitted through the glass and increase safety and security. The films help increase solar reflectivity, reject solar heat and reduce incoming ultraviolet light. 3M™ Window Films can significantly reduce the amount of heat coming through your windows to help lower utility costs.

3M Window Film can help you:

- Broad estimate electrical save 130-205 kwh/m2
- Reduce electricity bills as much as 10%
- Reduce temperatures as much as 9°F(5°C)
- Earn as many as 11 LEED points
- Increase your ENERGY STAR Portfolio Manager score
- Obtain utility rebates to help provide faster payback



LEED credit

- | | |
|---|---|
| • SS-8 (Light Pollution Reduction) | • MR 1.1-1.2 (Retain 75-95% of existing walls, floors and roof) |
| • EQ-7.1 (Thermal Comfort) | • EQ-8.1-8.2 (Daylight 75-95% of spaces) |
| • EA-1 (Optimized Energy Performance) | • MR 5.1-5.2 (Extracted, Processed & Manufactured Regionally) |
| • EQ-7.2 (Thermal Comfort Verification) | • ID (Environmental quality UV rejection) |



Solution for Energy Efficiency

Solar Control Films

Department of Energy (FEMP) Study Results

- **Maximum cost effectiveness rating**
- **Maximum probability of success**
- Window films are considered “top-tier technology”
- Window films have a typical **payback of:**

less than

3 years

CONSOL Energy Study Results

- Window films **pay back in <3 years**
- Window films **save up to:**

19 kWh/ft² on a **single pane**
or

12 kWh/ft² on a **double pane**

3M Solar Control Films have been proven to effectively improve energy savings

Department of Energy (DOE) Study

Background

In 2011, the US DOE completed a study on the top 50 commercially available energy conservation technologies. The technologies were ranked in these categories:

- Payback
- Probability of success
- Overall energy savings
- Technologies with less benefit that may be considered for specific targeted applications

Results

- Ranked as top-tier technology
- Fastest payback ranking available — approximately 3 years
- Highest probability of success ranking (based on customer acceptance, ease of retrofit, knowledge base of the technology and supply chain strength)

ConSol Study

Background

ConSol Energy and Environmental Solutions is a leading consulting firm for builders, government agencies, utilities and trade associations. ConSol utilized the US DOE-recommended software platform, Energy Plus, to calculate the effects of adding window film to a commercial building. The commercial building model used was the DOE recommended Energy Plus Commercial Building Benchmark Model. The study was completed in ICC Climate Zones 1, 2 and 3.

Results

This study further justified the DOE study with the following results:

- Single-pane glass showed paybacks in as short as 1.4 years
 - Double-pane glass showed paybacks in as short as 2.1 years
 - Annual energy savings as much as 19 kWh/sq ft (200 kWh/sq m) of glass
 - An 8.8% carbon emission reduction would occur if every home in California had window film installed
- After this study was published in 2011, the state of California updated its building code to include window films.²

Proven savings in every climate.

The US is broken down into eight different climate zones in accordance with building codes today — Climate Zone 1 being the hottest and Climate Zone 8 the coldest. 3M™ Window Films can save energy in all climate zones.

Climate Zone 1

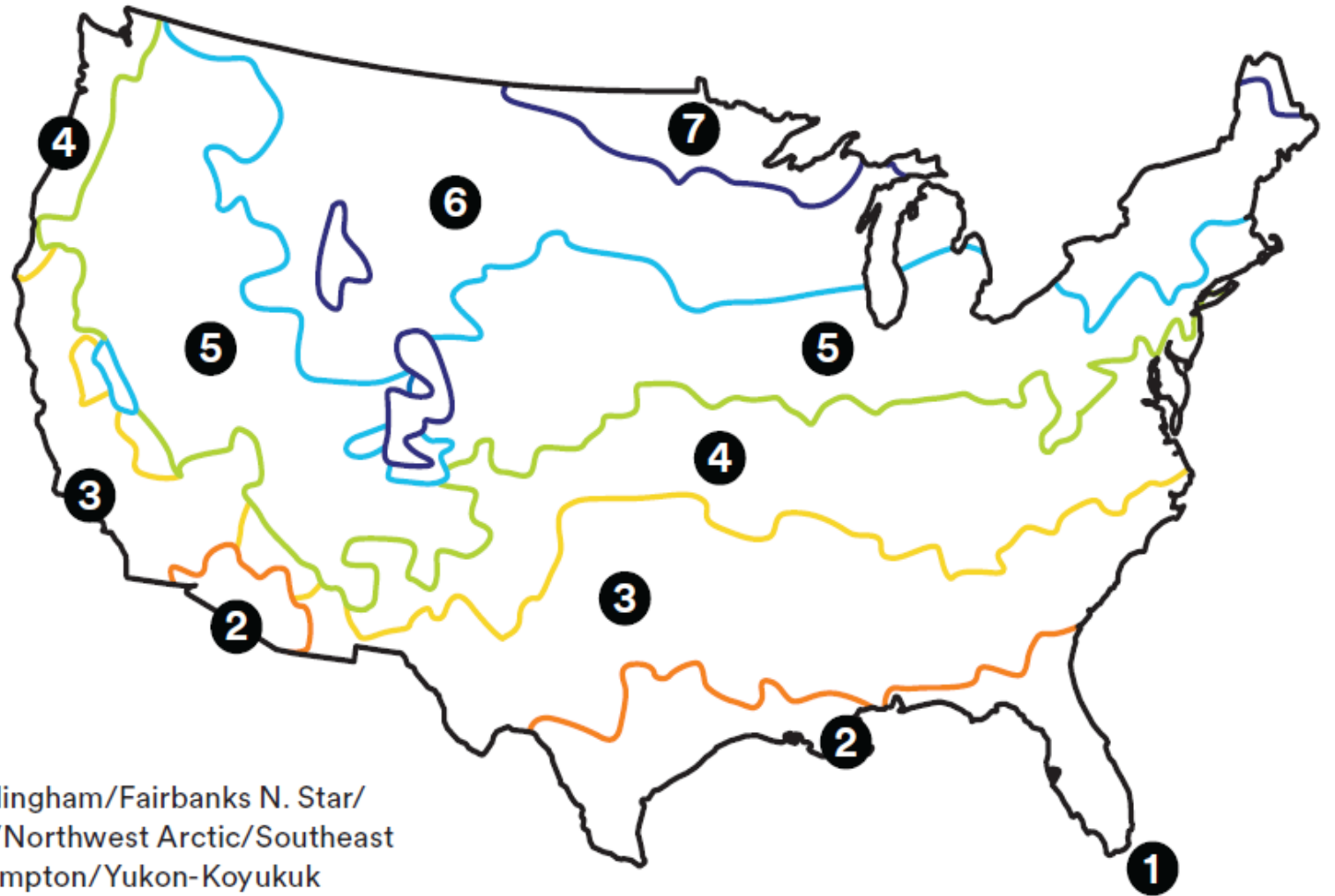
Includes: Guam/Hawaii/
Puerto Rico/Virgin Islands

Climate Zone 7

Includes: Alaska
Some Alaska boroughs are
also included in Zone 8

Climate Zone 8

Includes: Bethel/Dillingham/Fairbanks N. Star/
Nome/North Slope/Northwest Arctic/Southeast
Fairbanks/Wade Hampton/Yukon-Koyukuk





Climate Zone 1 | Case Study:

Mount Terrace Condominium, Hawaii Kai, Hawaii

Challenge: To retain a consistent look for the condominium complex while also finding ways to reduce the cost of managing the building.

Solution: 3M™ Sun Control Window Film Prestige 50 was chosen to obtain a maximum amount of energy rejection while maintaining the current look of the building. In addition, the 3M™ Prestige Series Window Film product was chosen because it has no metals and therefore no possibility of corrosion from the ocean air.

Results:

Payback — Less than 2 years

Savings — Greater than \$270,000 annually

Energy saved — 17 kWh/sq ft (180 kWh/sq m) of glass

Date completed — 2012



Climate Zone 2 | Case Study:

National Bank of Arizona, Phoenix, Arizona

Challenge: To retain the natural lighting while reducing both the heat it generated and glare on the banking floor. Minimizing disruption to normal banking operations was paramount.

Solution: 3M™ Sun Control Window Film Prestige Exterior 40 was installed because of its high infrared heat and harmful ultraviolet ray rejection. It is non-metallic and thus will not corrode, negating the need for edge sealing and minimizing installation time. Much of the project's cost was offset by an energy-efficiency rebate from Salt River Project (SRP).

Results:

Glass — Double pane tinted

Payback — Less than 5 years

Savings — Greater than \$25,000 annually

Energy saved — 12 kWh/sq ft (130 kWh/sq m) of glass

Date completed — 2011



Climate Zone 3 | Case Study:

Century Plaza Towers, Los Angeles, California

Challenge: To solve interior temperature imbalance and make the towers more energy efficient without drastically changing the look of the historic buildings.

Solution: 3M™ Sun Control Window Film Neutral 35, which offers high heat rejection and neutral light appearance, was installed. Much of the project's cost was offset by a rebate of \$116,000 from the Los Angeles Department of Water and Power.

Results:

Payback — Less than 1 year

Savings — Greater than \$200,000 annually

Energy Saved — Greater than 15 kWh/sq ft
(161 kWh/sq m) of glass

Date Completed — 2009



Climate Zone 4 | Case Study:

Fifth Third Center, Cincinnati, Ohio

Challenge: To control incoming light and heat in order to keep tenants comfortable, reduce energy costs and maintain the building's striking aesthetics.

Solution: Solar Tint, which had installed 3M™ Window Film in 1980, was chosen to update the facility's window film to further reduce energy costs by installing 3M™ Sun Control Window Film Neutral 20.

Results:

Payback — Less than 1 year

Savings — Greater than \$100,000 annually

Date Completed — 2000



Climate Zone 5 | Case Study:

1501 Clinton Avenue, Baltimore, Maryland

Challenge: To non-intrusively cut energy costs and increase interior comfort in a short time frame without affecting the tower's reflective modernism. Paramount was the issue of paying for up-front job costs on a limited budget.

Solution: 3M™ Sun Control Window Film Ceramic Series 30, which offers high clarity and outstanding heat reduction, was installed. To help pay for the project, 3M facilitated a custom rebate program with ICF International, the entity that supports Baltimore Gas and Electric's (BGE) energy efficiency programs.

Results:

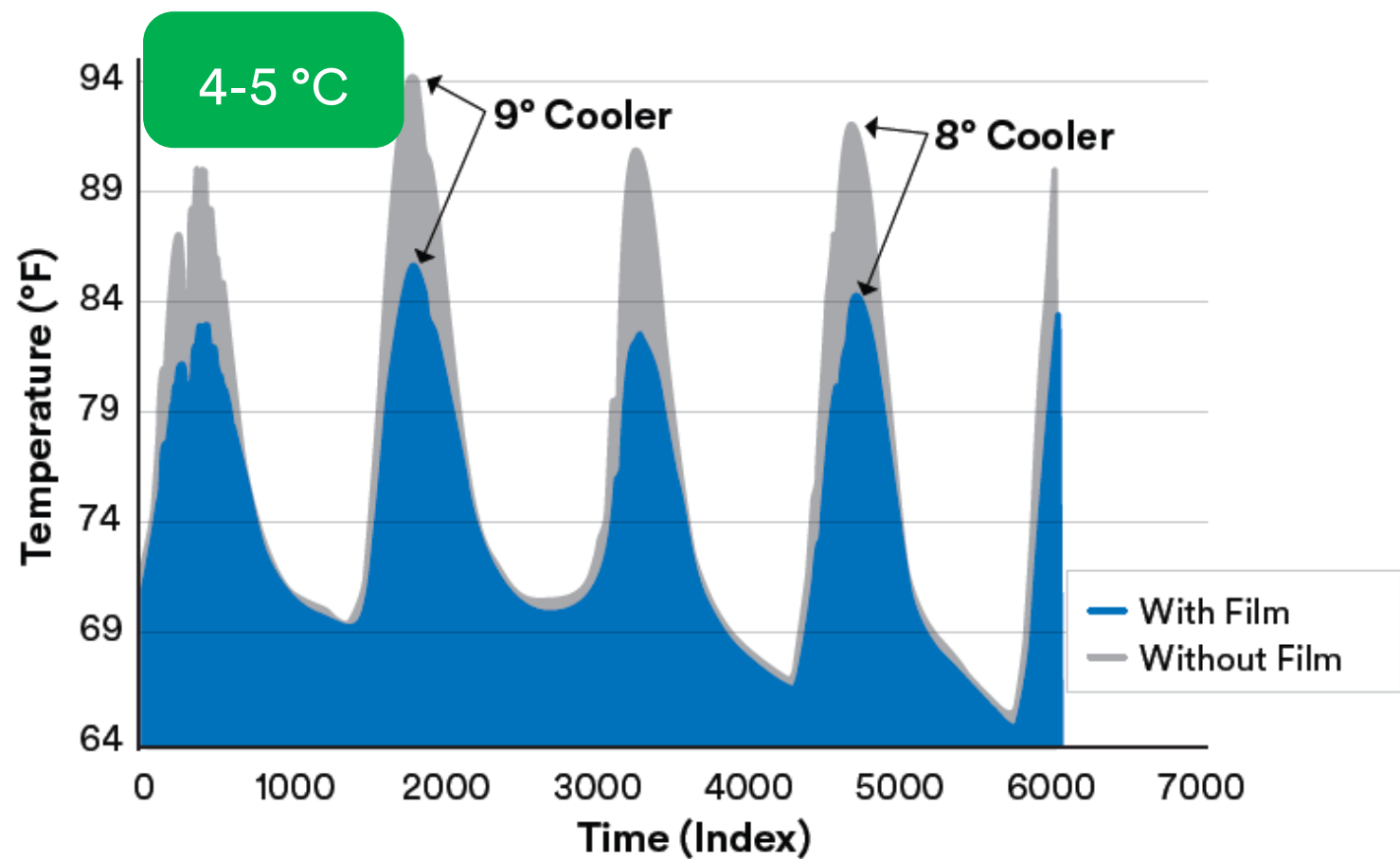
Payback — Less than 1 year

Savings — Greater than \$100,000 annually

Date completed — 2000

Comfortably cool.

3M™ Sun Control Window Film can reduce the temperature in direct sunlight by as much as 9° F, making your space much more usable and comfortable. The graph below shows a four-day temperature logging experiment conducted in an office building.

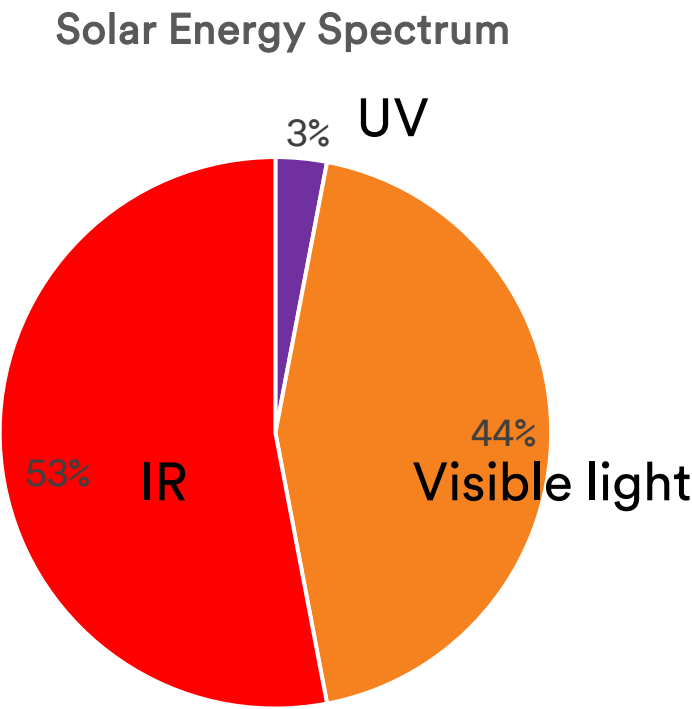
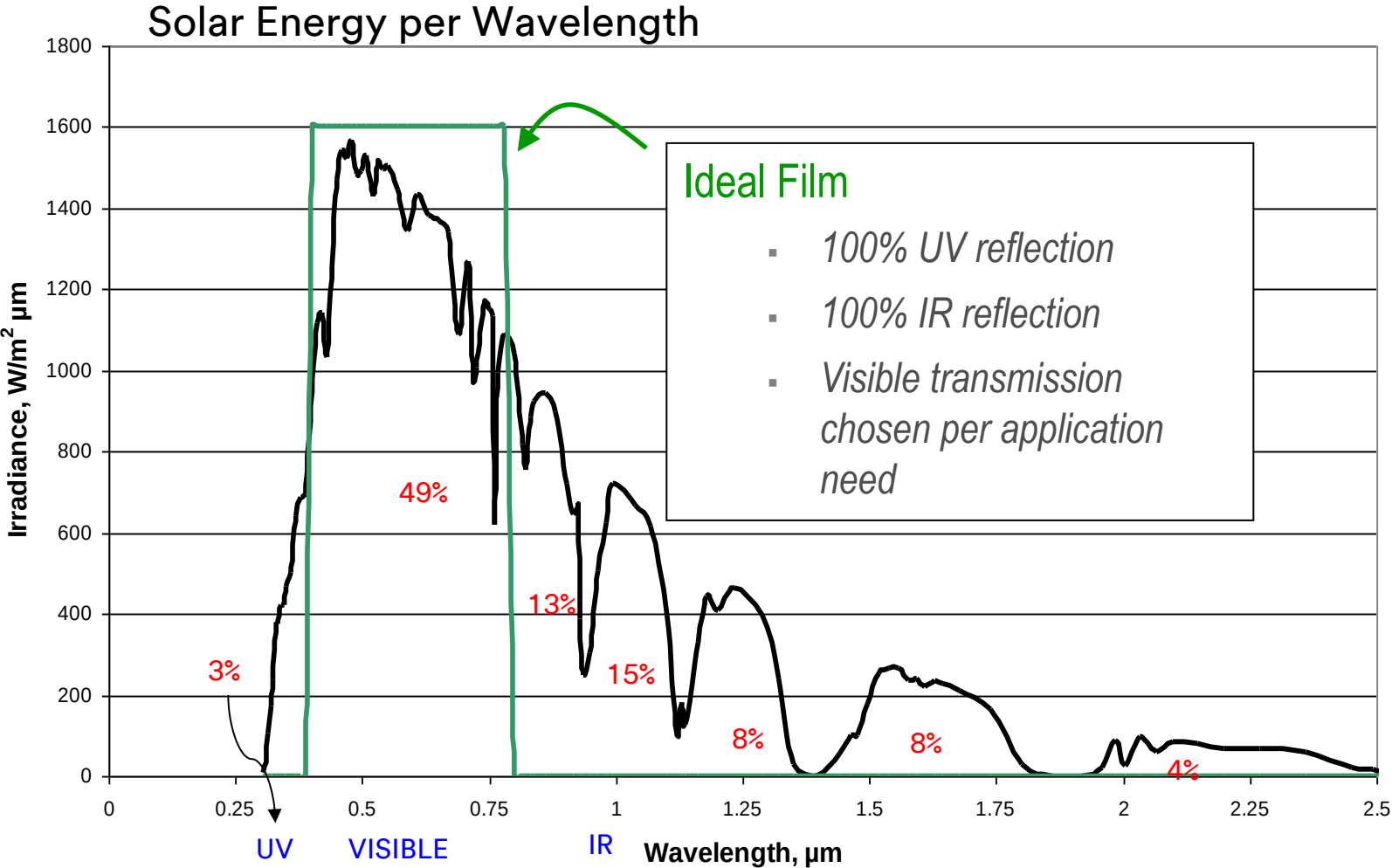


LEED Certification

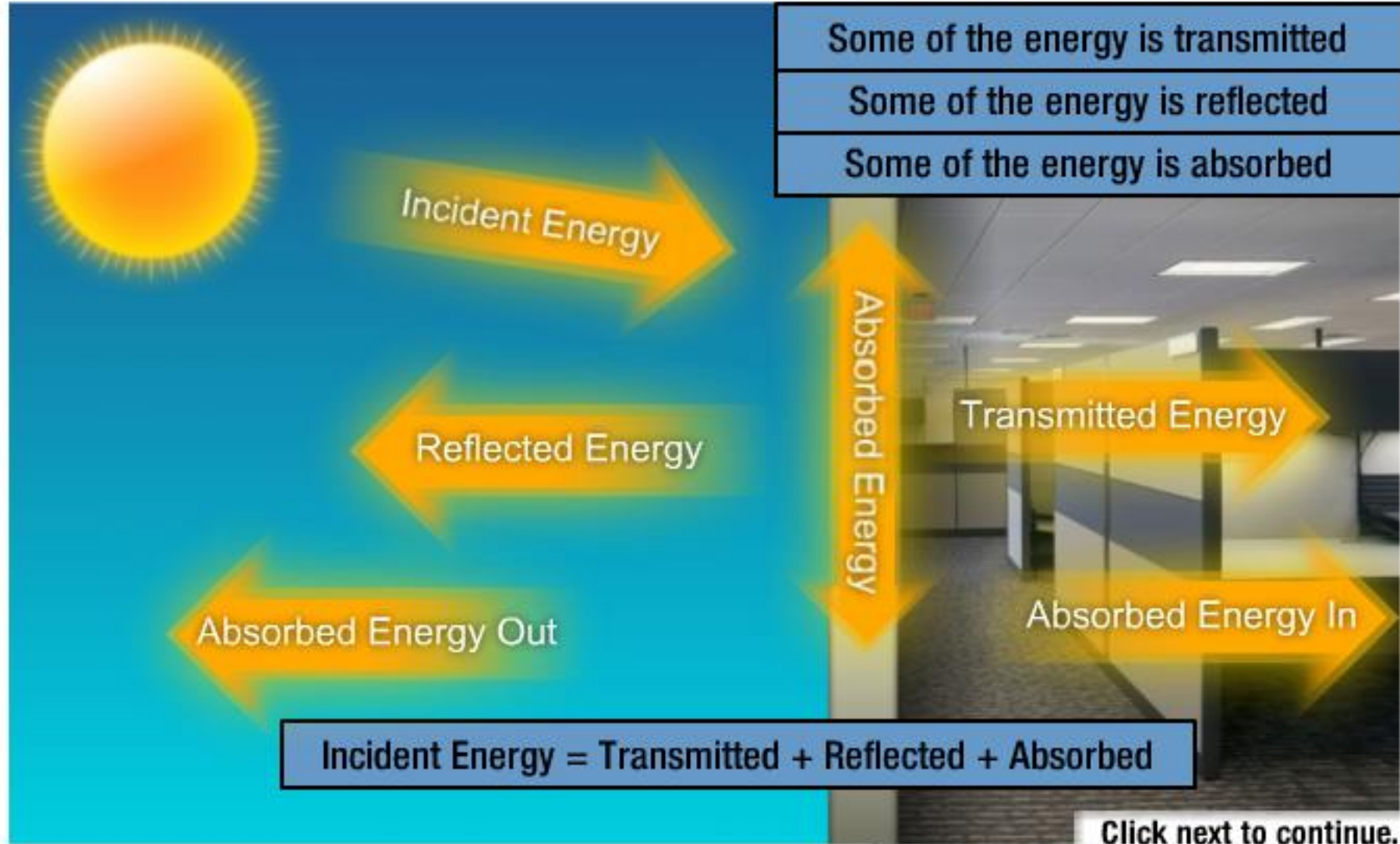
Window films may be used toward the following LEED credits:

▶ SS-8	▶ EQ-7.1
▶ EA-1	▶ EQ-7.2
▶ MR 1.1-1.2	▶ EQ-8.1-8.2
▶ MR 5.1-5.2	▶ ID

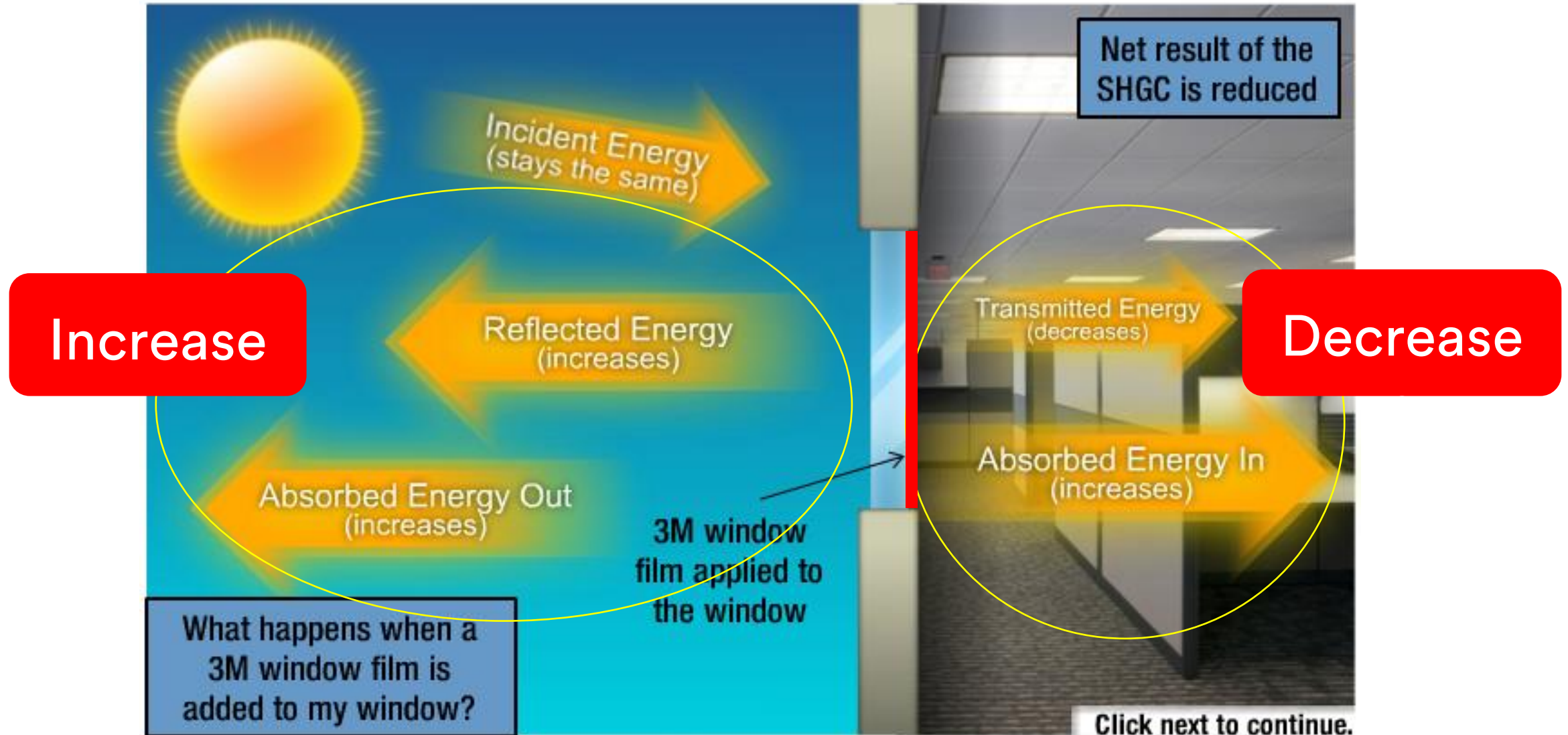
Window Film Basics



How Do Solar Films Work?

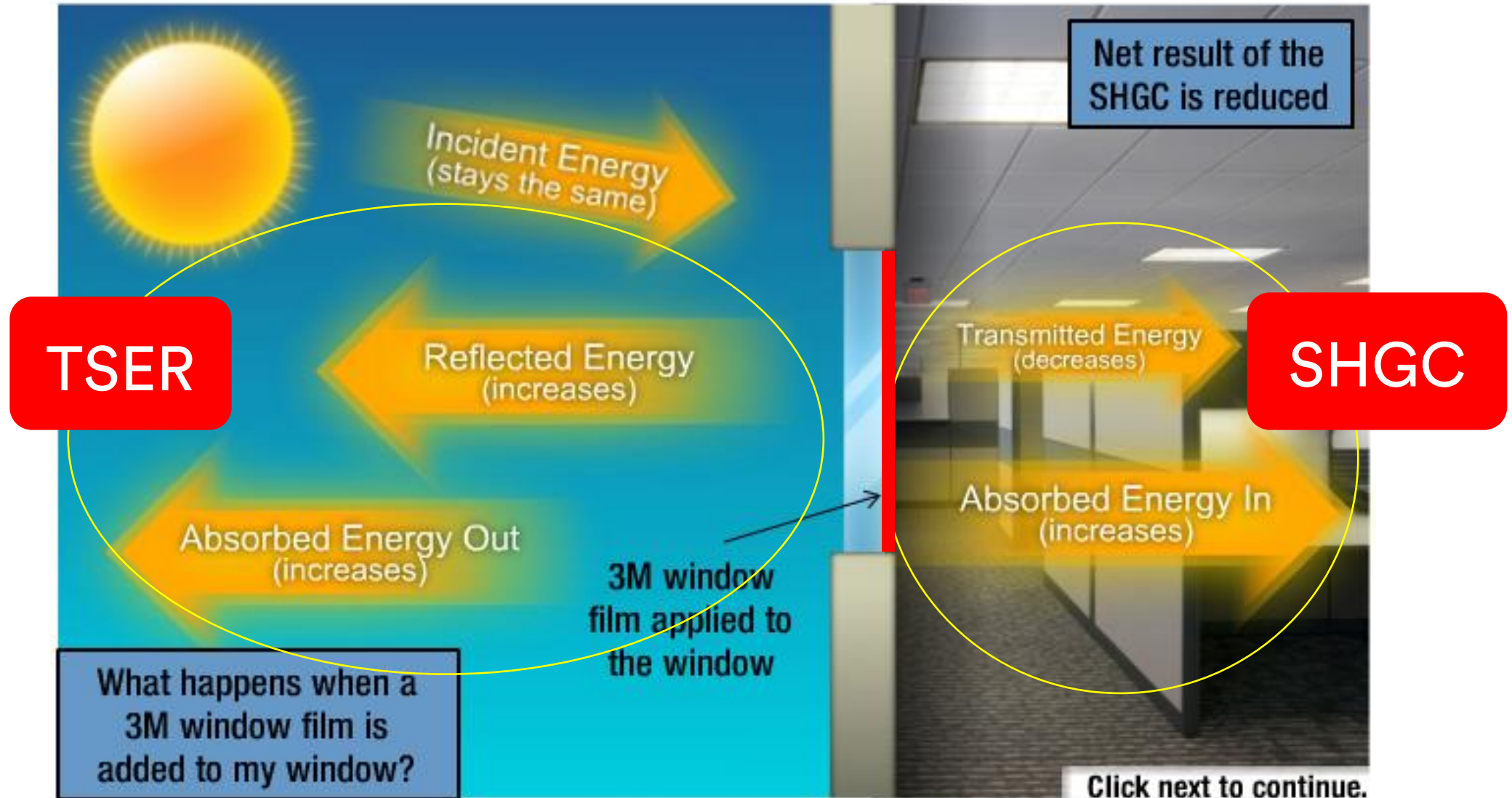


How Do 3M Solar Films Work?



3M After Installed Film

How Do 3M Solar Films Work?



3M After Installed Film

Benefit - Window Film

Window Film Can help :

1. Dramatic solar heat reduction.
2. Up to 99% UV block.
3. Better HVAC system management & lower air conditioning cost by reducing heat that comes into a building.
4. Save electricity cost.



Product specification : 3M CMIR70

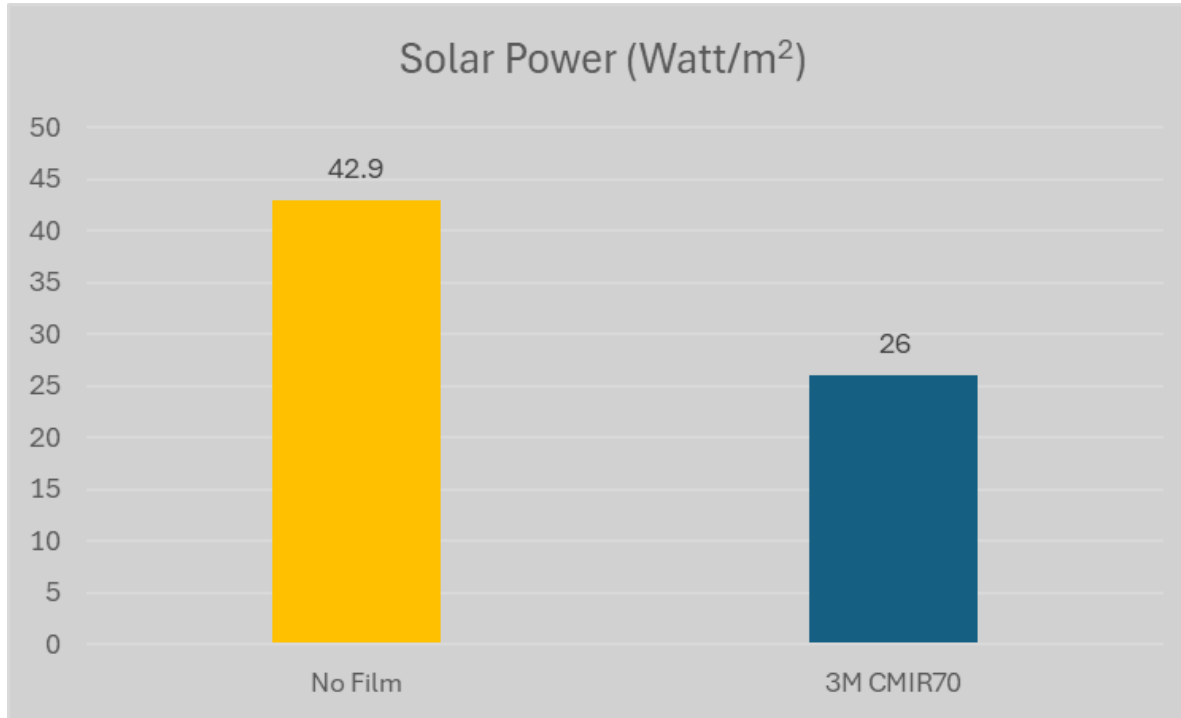
6mm Clear**	Film	Visible Light Transmittance	Visible Light Reflectance Exterior	Visible Light Reflectance Interior	UV Block	Total Solar Energy Rejected	Glare Reduction	Infrared Rejection****	Infrared Energy Rejection****
	Clear Glass	89%	9%	8%	34%	19%	NA	NA	NA
	Ceramic IR 5	6%	5%	5%	99.9%	63%	93%	95%	63%
	Ceramic IR 15	19%	5%	5%	99.8%	59%	78%	90%	61%
	Ceramic IR 25	31%	5%	5%	99.7%	57%	65%	90%	63%
	Ceramic IR 30	37%	6%	6%	99.9%	55%	58%	88%	62%
	Ceramic IR 35	45%	5%	6%	99.6%	52%	49%	85%	61%
	Ceramic IR 50	60%	7%	7%	99.6%	47%	32%	83%	60%
	Ceramic IR 70	78%	8%	8%	99.2%	41%	12%	78%	58%

Test result : Bangchak Project

Testing location



1) Solar energy measured by Solar Power Meter.



Solar energy measurement by Solar power meter



No Film



3M CMIR70

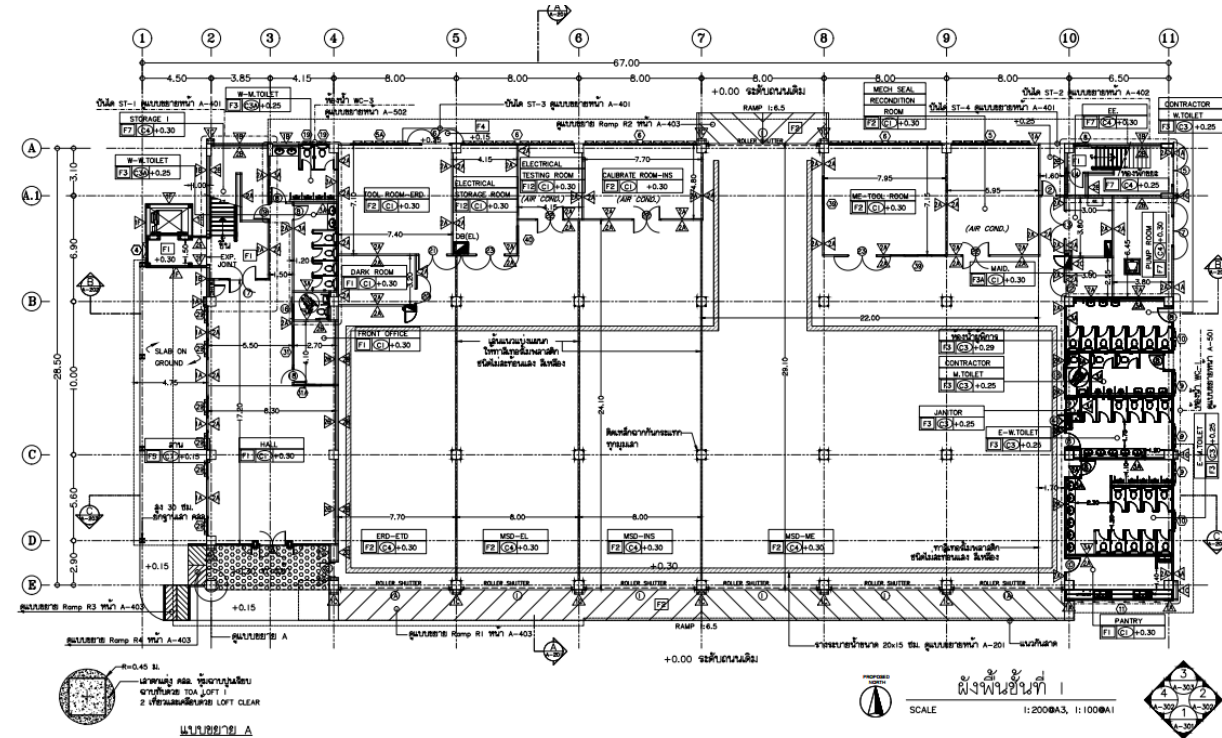
2) Payback Period calculated by Efilm software.

Data for calculation

Total glass area = 275.65 m²
Electricity cost = 4.50 THB/kWh.
Electricity bill = 1,754,065 THB/year.

Film cost

3M CMIR70 = 357,200 THB.



2) Payback Period calculated by Efilm software.

Building Energy Analysis Report

4/23/2025

Report generated by Efilm, an EnergyPlus based program.

Efilm version: v22.6.28

Simulation performed on: Wednesday, April 23, 2025

EnergyPlus version: 9.6

Project Information

Title: Bangchak Project

Address:

City/State/Zip: Bangkok,

Building Owner

Name:

Phone:

Address:

City/State/Zip:

Analyst

Name: Pichayapong Boonlue

Company: 3M Thailand Limited

Address: Latkrabang

City/State/Zip: Bangkok Thailand

Architect

Name:

Phone:

Address:

City/State/Zip:

Building Information

Building Type: Grocery Store/SuperMarket

Building Shape: Rectangular

Orientation: North

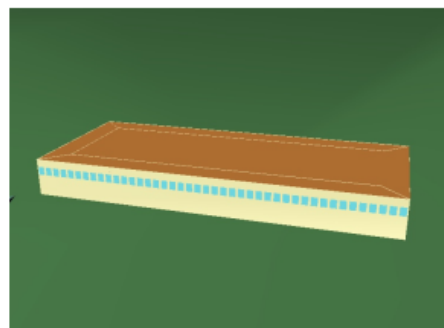
Number of Floors: 1

Floor-to-Floor Height: 7.096 m

Conditioned Area: 1,909.5 m²

Vintage: Post-1992

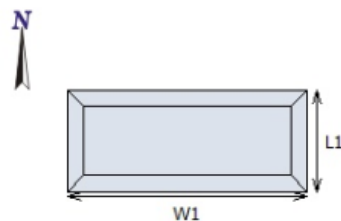
Category: Non-Residential



Dimensions

W1: 67 m

L1: 28.5 m



Results Summary

Option	Gas	Electricity	Total (Site)	Total (Source)	Savings
Energy Totals					
Base Case	78 GJ	380,644 kWh	1,449 GJ	4,420 GJ	
CMIR70	98 GJ	369,197 kWh	1,427 GJ	4,311 GJ	108 GJ
Energy/Area					
Base Case	40.964 MJ/m ²	199.342 kWh/m ²	758.6 MJ/m ²	2,314.61 MJ/m ²	
CMIR70	51.474 MJ/m ²	193.348 kWh/m ²	747.52 MJ/m ²	2,257.82 MJ/m ²	56.79 MJ/m ²
Peak Demand					
Base Case	35 kW	78 kW			
CMIR70	36 kW	74 kW			
Energy Cost					
Base Case	฿860	฿1,713,037	฿1,713,897		
CMIR70	฿1,081	฿1,661,522	฿1,662,603		฿51,294
Energy and HVAC Equipment Cost					
Base Case	฿860	฿1,713,037	฿1,713,897		
CMIR70	฿1,081	฿1,661,522	฿1,662,603		฿51,294

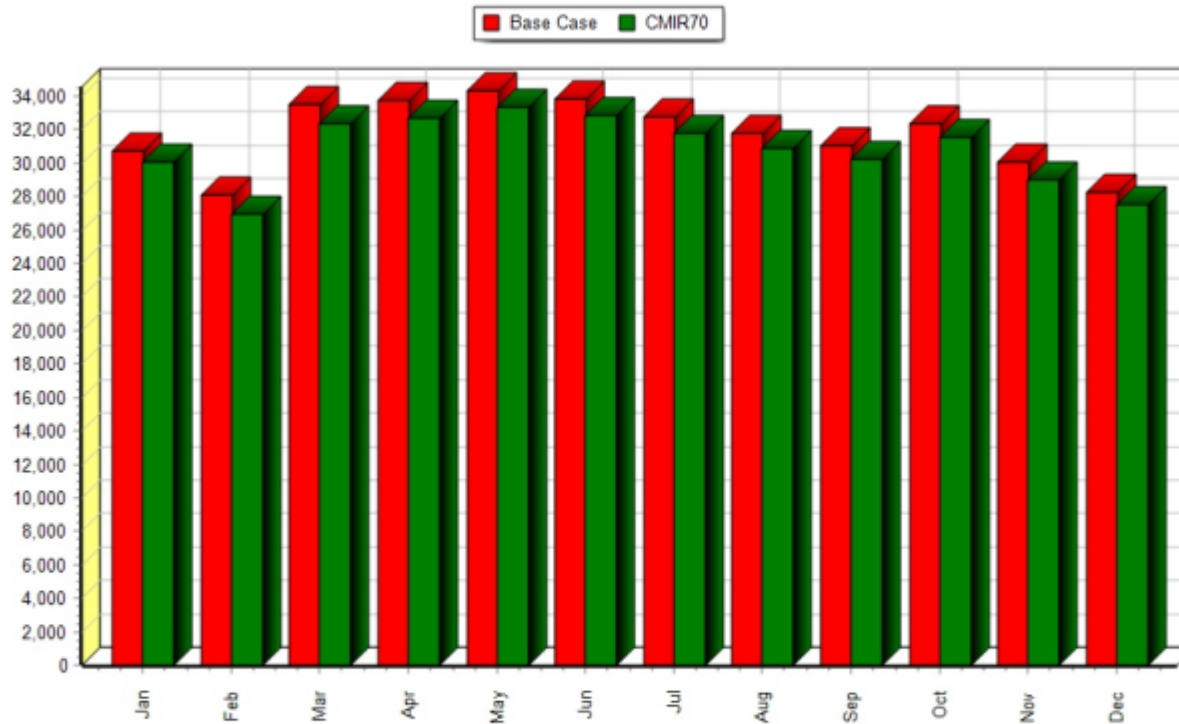
Simple Payback

Option	Project Cost	Simple Payback Period
Base Case	฿0	
CMIR70	฿357,200	6.96 years

1) Payback Period calculated by Efilm software.

Comparison Report

Monthly Electric Use [kWh]



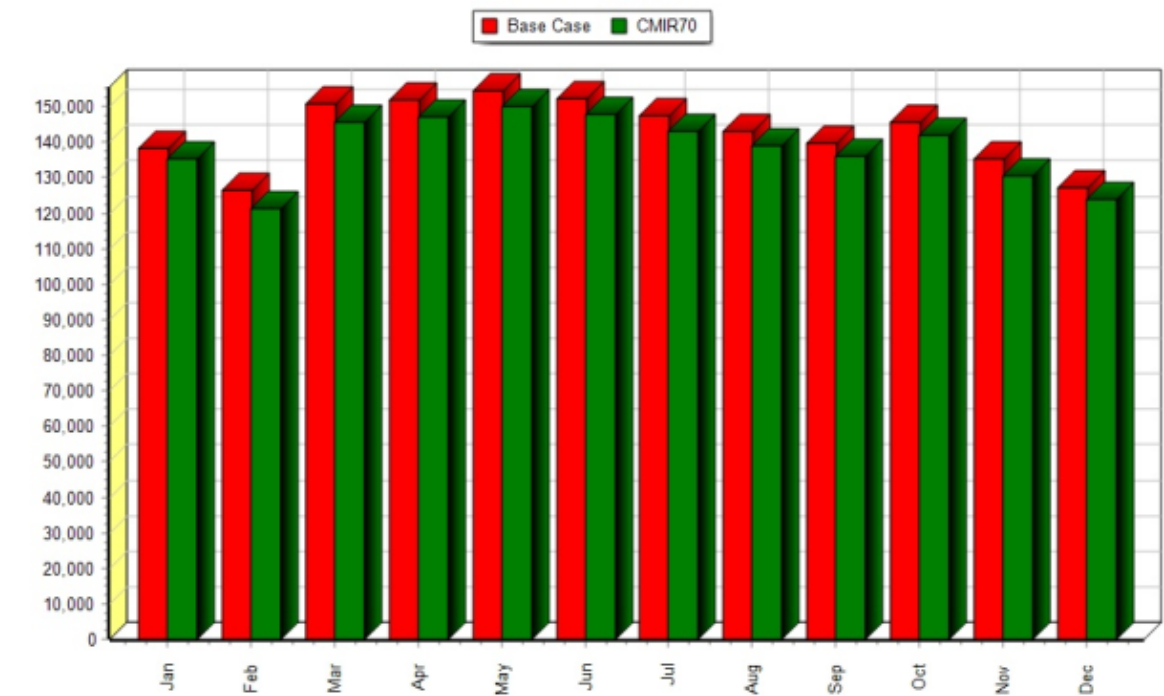
Electricity Saving (kWh)

CMIR70 = 11,447 kWh./year.

	01 Jan	02 Feb	03 Mar	04 Apr	05 May	06 Jun	07 Jul	08 Aug	09 Sep	10 Oct	11 Nov	12 Dec	Total
Base Case	30,716	28,112	33,515	33,784	34,331	33,868	32,773	31,768	31,079	32,405	30,062	28,231	380,644
CMIR70	30,034	26,956	32,348	32,679	33,336	32,848	31,792	30,925	30,210	31,520	29,035	27,514	369,198
CMIR70 - Savings	682	1,155	1,167	1,105	995	1,020	981	842	870	885	1,027	717	11,447

2) Payback Period calculated by Efilm software.

Monthly Electric Cost [?]



Electricity Cost Saving (THB)

CMIR70 = 51,515 THB/year.

	01 Jan	02 Feb	03 Mar	04 Apr	05 May	06 Jun	07 Jul	08 Aug	09 Sep	10 Oct	11 Nov	12 Dec	Total
Base Case	138,234	126,513	150,829	152,040	154,501	152,421	147,491	142,966	139,868	145,833	135,290	127,051	1,713k
CMIR70	135,163	121,314	145,578	147,069	150,024	147,828	143,076	139,175	135,954	141,851	130,667	123,823	1,662k
CMIR70 - Savings	3,071	5,200	5,251	4,971	4,477	4,592	4,415	3,791	3,915	3,981	4,623	3,228	51,515

2)Payback Period calculated by Efilm software.

Payback period

CMIR70 = 6.96 years.

Simple Payback

Option	Project Cost	Simple Payback Period
Base Case	?0	
CMIR70	?357,200	6.96 years

The background is a complex, abstract geometric pattern composed of numerous overlapping triangles. The color palette is warm, featuring various shades of orange, from light yellow-orange to deep, vibrant orange-red. The triangles vary in size and orientation, creating a dynamic and textured visual effect.

Thank you